MD. MIZANUR RAHAMAN

Master's Student (Final Year), Department of Civil Engineering
The University of Tokyo, Japan

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Research Area: Structural Health Monitoring (SHM), AI for Infrastructure Inspection, Computer Vision, BIM & Digital Twin Systems

EDUCATION

The University of Tokyo, Japan
Master of Engineering (M.Eng.) in Civil Engineering
Field: Infrastructure Development and Management
Laboratory: Construction Management and Infrastructure Systems
Supervisor: Associate Professor Pang-jo Chun
Oct 2025 – Sep 2027 (Expected)
ADB-JSP Scholarship Recipient
Research Focus: AI-enabled bridge damage detection, severity classification, and BIM-integrated visualization for structural health monitoring.

Khulna University of Engineering & Technology (KUET), Bangladesh
B.Sc. in Building Engineering and Construction Management
CGPA: 3.55/4.00 (Ranked 13th out of 41)
Feb 2014 – Apr 2018
Undergraduate Thesis: Influence of Recycled Waste Glass as Fine Aggregate in Brick Aggregate Concrete

RESEARCH EXPERIENCE

Graduate Researcher
Construction Management and Infrastructure Systems Laboratory
The University of Tokyo, Japan
Oct 2025 – Present

- Developing a three-stage AI-based framework for automated bridge damage inspection using computer vision and BIM integration.
- Applying YOLO-based segmentation models for crack and corrosion detection in steel bridge inspection images.
- Investigating pixel-to-metric digital scaling methods for automated severity quantification.
- Developing BIM localization workflow using Python and Dynamo for spatial mapping of damage information.
- Integrating inspection data into damage-aware digital twin environments for infrastructure monitoring.

PUBLICATIONS

Maleque, M. S. E., Rahaman, M. M., & Sobuz, M. H. R. (2020).
“Rheological and Mechanical Properties of Recycled Waste Glass Concrete as Partial Replacement of Fine

Aggregate.”
5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020), KUET, Bangladesh.

PROFESSIONAL EXPERIENCE

Senior Structural Engineer
Sthapona Consultants Limited, Bangladesh
Mar 2019 – Sep 2025

- Structural analysis and design of reinforced concrete and steel structures.
- BIM-assisted structural coordination and engineering assessment.
- Experience in retrofitting, structural auditing, and construction quality evaluation.

Quality Control Engineer
Sthapona Consultants Limited, Bangladesh
May 2018 – Mar 2019

- Construction quality inspection and material testing for RCC and steel structures.
- Coordination of field engineering and construction monitoring activities.

TECHNICAL SKILLS

Programming & AI: Python, Machine Learning, Computer Vision, Deep Learning (*PyTorch, OpenCV, Ultralytics YOLO*)

BIM & CAD: Revit, Navisworks, Autodesk BIM 360, AutoCAD

Structural Engineering Software: ETABS, SAFE, SAP2000, STAAD.Pro, Tekla Structural Designer, IDEA StatiCa

Miscellaneous: OriginPro, Zotero, MS Excel, MS Office

RESEARCH INTERESTS

- Structural Health Monitoring (SHM)
- AI for Infrastructure Inspection
- Computer Vision in Civil Engineering
- BIM & Digital Twin Systems
- Bridge Damage Detection and Quantification
- Smart Infrastructure Monitoring

AWARDS & SCHOLARSHIPS

- Asian Development Bank – Japan Scholarship Program (ADB-JSP), The University of Tokyo, 2025–2027
- Undergraduate Technical Merit Scholarship, KUET, 2014–2018

STANDARDIZED TEST SCORES

IELTS Academic: Overall 7.0 (L: 7.5, R: 8.0, W: 6.5, S: 6.0)
GRE General: 304 (Q: 162, V: 142, AWA: 3.0)